

no_root_squash: By default, root on an NFS client is treated as user `nfsnobody` by the NFS server. That is, if root attempts to access a file on a mounted export, the server will treat it as an access by the user `nfsnobody` instead. This is a security measure that can be problematic in scenarios where the NFS export is used as / by a diskless client and root needs to be treated as root. To disable this protection, the server needs to add `no_root_squash` to the list of options set for the exports in `/etc/exports`.

```
/myshare diskless.example.com(rw,no_root_squash)
```

Warning: Note that this particular configuration is not secure, and would be better done in conjunction with Kerberos authentication and integrity checking.

Configuring an NFS export

In this example, we follow the steps on how to share a directory that is IP-based, with NFS. The directory `/myshare` is on `server0` and will be mounted on the client `desktop0` system.

1. Configure an IP based NFS share on `server0` that provides a newly created share directory `/nfsshare` for the `desktop0` machine with read and write access.
 - Start the NFS service on server with the `systemctl` command:

```
# systemctl start nfs-server
```

- Enable the NFS service to start at boot on `server0`:

```
# systemctl enable nfs-server
```

- Create the directory `/myshare` to share it with NFS on the `server0` system.

```
# mkdir /myshare
```

- Export the `/myshare` directory on `server0` to the `desktop0` client as a read and write enabled share. To do that, add the following line to the `/etc/exports` file on `server0`:

```
# echo "/myshare desktop0(rw)" >> /etc/exports
```

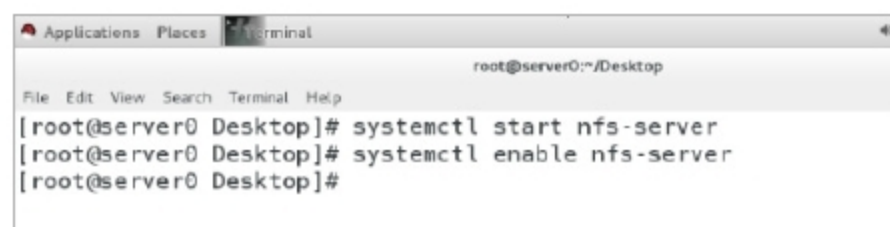


Figure 1: NFS server service has been started and enabled



Figure 2: Directory `myshare` has been created

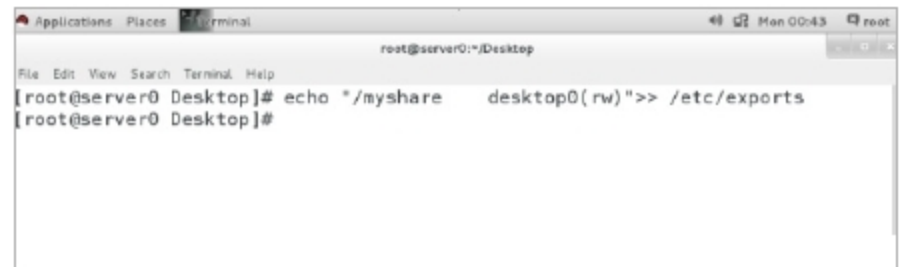


Figure 3: Entry is made in the `/etc/exports` file



Figure 4: Changes have been applied by using the command `exportfs`

- After the changed `/etc/exports` file has been saved, apply the changes by executing the following command:

```
# exportfs -r
```

- The NFS port `2049/tcp` for `nfsd` must be open on the server. To configure `firewalld` to enable access to the NFS exports immediately, run the following command:

```
# firewall-cmd --permanent --add-service=nfs
```

- Reload the `firewalld` rules so that the new rule gets applied, as follows:

```
# firewall-cmd --reload
```

2. Mount the NFS export from the `server0` system on the `/mnt/nfsshare` mount point on `desktopX` permanently.

- Create the mount point `/mnt/nfsshare` on the `desktop0` system.

```
# mkdir /mnt/nfsshare.
```

- Create the required entry in `/etc/fstab` to mount the exported NFS share on the newly created `/mnt/nfsshare` directory on the `desktop0` system permanently.

```
# echo "server0:/myshare /mnt/nfsshare nfs defaults 0 0" >> /etc/fstab.
```



Figure 5: Firewall has been updated and reloaded

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